Q1:

a) by using the formula: $P = \rho g h$

$$\rho_{water}\, g\, h_{water} = \rho_{Hg}\, g\, h_{Hg}$$

$$1.00 \times h_{water} = 13.7 \times h_{Hg}$$

$$h_{Hg} = \frac{1}{13.7} \times 12\,m \approx 0.876\,m$$

b) Pressure : $155 + 435 + 122 = 712$ mm

$\therefore$ mole fraction
$$\begin{cases} CH_4 : \frac{155}{712} = 0.218 \\ N_2 : \frac{435}{712} = 0.611 \\ O_2 : \frac{122}{712} = 0.171 \end{cases}$$

c) by using formula: $PV = nRT$

$$\therefore\ n = \frac{PV}{RT}$$

$T = 114°C = 114 + 273.15 = 387.15K$.

$V = 368mL = 0.368L$.

$m = 1.325g$

$P = 946\,mmHg = \frac{946}{760} = 1.245\,atm$

$$n = \frac{0.458}{31.76} = 0.014\,mol$$

$$M = \frac{m}{n} = \frac{1.325g}{0.014\,mol} = 91.9g.$$

M_{emp}: NO_2 ($N:O = 1:2$)

$$\therefore\ 14.01 + 2(16.00) = 46.01g$$

$$91.9g \div 46.01g \approx 2$$

$\therefore (NO_2)_2 = N_2O_4$. (molecular formula)

Q_2 a) by using $\frac{P_1}{T_1} = \frac{P_2}{T_2}$

$$T_2 = T_1 \times \frac{P_2}{P_1}$$

$$= 298 \times \frac{2}{0.8} = 298 \times 2.5$$

$$= 745K$$

$745K = 471.85°C$.

b) $T = 27°C = 300K$ $\quad P = 0.5$ atm

$M_{SF_6} = 32.06 + 6(19.00) = 146.06 g/mol$.

by using $\rho = \frac{PM}{RT}$

$$= \frac{(0.5)(146.06)}{(0.0821)(300)} \approx \frac{73.03}{24.63} = 2.97 g/L$$

c) $\therefore V = \frac{nRT}{P}$

$\therefore$ linear, R. T, P are consistent.

Q_3:

a) $T = 10°C = 10 + 273.15 = 283.15K$

$M_{CO_2} = 44.01 g/mol = 0.044 kg/mol$

by using $u_{rms} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3(8.314)(283.15)}{0.044}} = 401 m/s$

b)

i : C

ii : C

iii : A<B<C.

Q4:

Q5:

$2A + B \rightarrow$ product.

	A	B	Rate
1	0.15	0.1	0.022
2	0.3	0.1	0.044
3	0.3	0.2	0.022

a) Rate $= k[A]^m [B]^n$

$$\frac{R_2}{R_1} = \frac{0.044}{0.022} = 2 = \left(\frac{0.3}{0.15}\right)^m = 2^m$$

$\therefore m = 1$

$$\frac{R_3}{R_2} = \frac{0.022}{0.044} = 0.5 = \left(\frac{0.2}{0.1}\right)^n = 2^n$$

$\therefore n = -1$

$$\text{Rate} = k[A]^1[B]^{-1} = k\frac{[A]}{[B]}$$

use Trial 1 data : $0.022 = k\frac{0.15}{0.1}$

$$k = 0.0147 M.$$

b)

$k = 6 \times 10^{-2} \text{ min}^{-1}$

$$\ln\frac{[A]_0}{[A]} = kt$$

$$\ln\frac{0.75}{0.25} = (6\times10^{-2})t$$

$$t = \frac{1.0986}{0.06} = 18.31 \text{ min}.$$

Q6:

a)

$$\ln \frac{[A]_0}{[A]} = kt$$

$$\ln \frac{0.5}{0.35} = 12k$$

$$k = 0.0297\ min^{-1}$$

$$\ln \frac{[A]_0}{[A]} = kt$$

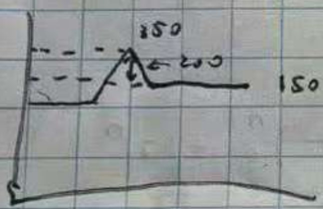
$$\ln \frac{[0.5]}{[0.15]} = 0.0297t$$

$$t = \frac{1.203}{0.0297} = 40.5min.$$

b) $\Delta H = E_{forward} - E_{reverse}$

$= 350 - 150 = +200$

∴ endothermic.



Q7

formula $\begin{cases} 1 \\ 2 \text{ (inverse)} \\ 3 \end{cases}$ -> $2ClF + 2F_2 \rightarrow 2ClF_3$

$\Delta H = 167.4 - 341.4 - 43.4$

$= -217.4kJ.$

$-217.4 \div 2 = -108.7kJ$ for $ClF + F_2 \rightarrow ClF_3$

b) $q = mC\Delta T$

$= 28.6 \times 4.186 \times 56.3$

$= 6.74kJ.$

Q8:

$q_{cu} = -q_{H_2O}$

$q_{H_2O} = mc\Delta T = (50)(4.18)(25-20)$

$= 1045 J = q_{cu}$

$\Delta T_{cu} = 100 - 25 = 75 k$

$$C_{cu} = \frac{q}{\Delta T} = \frac{1045}{75} = 13.9 J/k$$

b) $2C + 3H_2 + \frac{1}{2}O_2 = C_2H_5OH$.